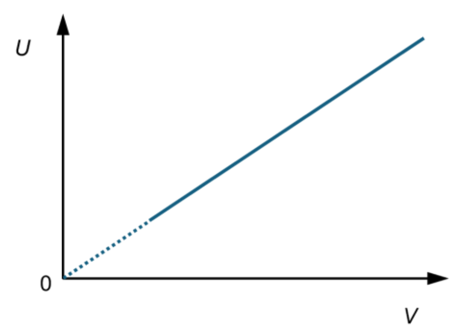
Qn	Suggested Solutions
3(a)	The internal energy U of a system is the sum of a random distribution of potential and kinetic energies of the atoms/molecules/particles in the system.
(b)	⅓: molecules move randomly in three dimensions (not one) so the mean square speed is any one direction is ⅓ of the mean square speed
	$<c^2>$: molecules have different/a range of speeds so take average of the square of speeds
(c)	$p = \frac{1}{3}\rho\langle c^2\rangle$ $p = \frac{1}{3}\left(\frac{Nm}{V}\right)\langle c^2\rangle$ $pV = \frac{1}{3}Nm\langle c^2\rangle$ $\frac{3}{2}pV = \frac{1}{2}Nm\langle c^2\rangle = E_k$
	Using the ideal gas equation of state, $\frac{3}{2}NkT = E_{\mathrm{k}}$
	For an ideal gas, there is no intermolecular forces of attractions so it has no microscopic potential energy. $U = E_{\mathrm{k}}$ Hence, $U \propto T$
(d)(i)	$pV = nRT$ $n = \frac{pV}{RT}$ $= \frac{(2.0\times10^5)(0.26)}{(8.31)(273.15+20)}$ $= 21\text{ mol}$
	$N = nN_A$ $= (21.34)(6.02\times10^{23})$ $= 1.3\times10^{25}$
(ii)	$U = \frac{3}{2}NkT$ $= \frac{3}{2}(1.285\times10^{25})(1.38\times10^{-23})(273.15+20)$ $= 7.80\times10^4\text{ J}$
(e)	 Since $U \propto T$ and for constant p, $V \propto T$, therefore $U \propto V$ and it will be a straight line.

(f)	$\Delta U = Q + W$ For a constant pressure process with an increase in temperature, negative work is done on gas. $Q = \Delta U - W$, where W is negative.
	At constant volume, no work is done on gas. $Q = \Delta U$
	Hence, the heat supplied to raise the temperature by 1 kelvin ($C = Q / \Delta T$) will be higher for the constant pressure process. C_p will be higher than C_v.